MIL-C-46270B(MU)
14 March 1974
SUPERSEDING
MIL-C-46270A(MU)
9 March 1972

MILITARY SPECIFICATION

CARTRIDGE, DELAY: M71, ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cartridge, Delay: M71, Assembly, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MS-28741	- Hose Assembly, Detachable End-Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Armament Command

D8593162	- Cartridge, Delay: M71, Assembly
C8845190	- Carton, Packing, Ammunition for Cartridge, Delay, M71
C8845191	- Box, Packing, Ammunition for Cartridge, Delay, M71
IEL-8593162	- Inspection Equipment List for Cartridge, Delay: M71, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the delay charge, primer, igniter or propellant or derangement of components.

3.4 Primer sensitivity.-The primer shall fire when drop tested at an energy level of 26 inch ounces in a sectioned cartridge case, void of delay charge, igniter and propellant charge.

3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

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3.5.2 Ballistics.-The cartridge when tested in an M12A1 or M33 Initiator shall produce pressures and delay times at the temperature specified in Table I.

TABLE I

Temperature	Pressure (PSI) 1/ Minimum	Maximum	Delay Time, Seconds (See 6.2)
-65° to -70°F	1300	2200	1.3 max.
65° to 75°F	1300	2200	1.0 ± 0.1
160° to 165°F	1300	2200	0.80 min.

1/ Pressue values shall be produced in a one cubic inch pressue station at the end of 3 feet of hose (Size No. 4) in accordance with MS-28741.

3.6 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

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4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 cartridges for primer sensitivity, 5 cartridges for sealant tests (to be selected from primer sensitivity sample), 30 cartridges for ballistic tests, 12 for vibration tests (to be selected from ballistics sample), and 10 sets of unassembled components for dimensional and visual examination and tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE II

Temperature	Vibrated	Not Vibrated
-65° to -70°F	4	6
65° to 75°F	4	6
160° to 165°F	4	6

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-An ammunition lot of cartridges shall consist of an identifiable group of cartridge assemblies consisting of:

a. Components produced by a homogeneous process.
b. Delay elements from not more than one lot.
c. Primers from not more than one lot.
d. Igniters from not more than one lot.
e. Propellant from not more than one lot.

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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables III and IV. For an individual defect (single characteristic) the criteria in Table III shall be used. For total defects within a major or minor class, the criteria in Table IV shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber "O" rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE IV

	Total Allowable Defects Per Sample Size			
Sample Size for Major Defects - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

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4.3.2.1 Sealing Disc, Dwg. C8594077, covering a detail of Dwg. D8593162.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Diameter, min.	Gage
102. Thickness	Gage
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Varnish missing or inadequate	Visual

4.3.2.2 Case, Cartridge, Dwg. C8595202, part No. 8595392, covering a detail of Drawing D8593162.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of body, min.	Gage
104. Diameter of head, min.	Gage
105. Thickness of base	Gage
106. Thickness of wall	Gage
107. Location of datum diameter	Gage
108. Protective finish inadequate or defective	Visual
109. Burr or sharp corner	Visual
110. Foreign matter	Visual
MINOR	
201. Diameter of head, max.	Gage
202. Diameter of body, max.	Gage
203. Total length	Gage
204. Radius or chamfer missing or incorrect	Visual
205. Surface finish improper	Visual

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4.3.2.3 Delay Element M8 Dwg. C8593163 covering a detail of Dwg. D8593162 .

Defect	Method of Inspection
CRITICAL	
1. Mixed delay elements	Visual
2. Flash hole missing or obstructed	Visual
3. Primer uncrimped or seal missing	Visual
4. Marking missing, incorrect or illegible	Visual
5. Delay charge not visible through flash hole	Visual
6. Metal defective	Visual
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.3.2.4 Cartridge, Delay, M71, Drawing D8593162.

CRITICAL	
1. Any part missing	X-ray & Visual 1/
2. Improper assembly	X-ray & Visual 1/
3. Mixed delay element or propellant	X-ray & Visual 1/
4. Metal defective	Visual
5. Crimp missing or incorrect	Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Cartridge profile failure	Gage
102. Diameter of body, max.	Gage
103. Diameter of head, max.	Gage
104. Location of datum diameter	Gage
105. "O" Ring split, cracked or cut	Visual
106. Protective finish inadequate or defective	Visual
107. Foreign matter	Visual
108. Burr	Visual

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Total length	Gage
202. Case damaged	Visual
203. Varnish missing or inadequate	Visual

1/ The cartridge shall be x-rayed with the long axis of the case perpendicular to the rays. The x-ray negatives properly identified, shall be forwarded to the procuring activity.

4.3.2.5 Carton, Packing, Ammunition for Cartridge Delay M71, Dwg. C8845190.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separators or fillers missing	Visual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual

4.3.2.6 Box, Packing, Ammunition for Cartridge, Delay M71, Drawing C8845191.

CRITICAL	
None defined	
MAJOR	
None defined	

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Test methods and procedures.

4.3.3.1 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in Table III and IV and MIL-C-60539. Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge cases shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.3.3.2 Vibration.-Twelve of the 30 assembled x-rayed cartridges of the initial production sample shall be subjected to vibration in accordance with 3.3. After vibration the 12 vibrated cartridges shall be x-rayed as directed in note 1/ of 4.3.2.4. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the method of charging the cartridge or in the lot of primers, 12 additional cartridges shall be selected and tested as indicated herein. The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.3.3.3 Airtightness.-Each cartridge shall be subjected to an airtightness test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the cartridge. The method and procedure for performing the airtightness test shall be by use of equipment listed on IEL 8593162.

4.3.3.4 Primer sensitivity.-A sample of 50 assembled cartridges from the initial production sample and 25 assembled cartridges from each subsequent production lot shall be tested for sensitivity. Failure of one cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot. Remove and discard the charged delay body and all powder charges from the test cartridges

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4.3.3.4 (Cont'd)
by sectioning the cartridge case around its perinhery, one inch, maximum, above the base. Drop test 50 sectioned cartridges from the initial production sample and 25 sectioned cartridges from each production lot (all void of charged delay body and powder charges) at 13 inches, using a steel ball 15/16 inches in diameter, weighing 1.94 ounces plus or minus 0.02 ounces, in a primer sensitivity machine with a cartridge folder approved by the procuring activity.

4.3.3.5 Sealant application.-Five cartridges from each production lot, (from primer sensitivity sample), shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot. The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length (360°) of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

TABLE V

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
160° to 165°F	10

4.3.3.6 Hardness.-Ten units from each heat treated lot shall be selected for hardness tests. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components. A hardness testing machine shall be used to determine the hardness values specified on the applicable drawing.

4.3.3.7 Functional.-The following tests shall be performed by the Government:

4.3.3.7.1 Ballistics.-Thirty test cartridges from each production lot shall be grouped and temperature conditioned in accordance with 4.3.3.7.1.1 and Table V. Failure of any cartridge to comply with the requirements of 3.5.2 shall be cause for rejection of the lot. The sample for ballistic tests shall be grouped and temperature conditioned in accordance with 4.3.3.7.1.1 and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this teme limit is exceeded, the sample unit shall be reconditioned in accordance with 4.3.3.7.1.1.

4.3.3.7.1.1 Temperature conditioning.-A standard inert cartridge containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 8593162.

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5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.-Packing and marking shall be in accordance with Drawing C8845190 and C8845191.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Delay time is defined as the time that elapes between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Loaded Assembly Hazard Information:
Explosive Hazard Classification: Class 3
Storage Compatibility: Group B
Department of Transportation (DOT) Hazard Class C
DOT Marking - Explosive Power Devices - Class C
Handle Carefully - (Keep Fire Away)

6.4 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive, consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

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Preparing Activity:

Army - MU

Project No. 1377-A572

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DOCUMENT IDENTIFIER AND TITLE

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MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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b. Cavitation or porosity is not permitted at the interface of